Dashboard BOM & Pinout

Component	Interface	Signal	ESP32 GPIO	Notes
MAX7219 7-Segment Display	SPI	DIN	GPIO23	Shared SPI MOSI
	SPI	CLK	GPIO18	Shared SPI SCK
	SPI	CS	GPIO5	Dedicated Chip Select
	Power	VCC	5V	
	Power	GND	GND	
nRF24L01+	SPI	MOSI	GPIO23	Shared SPI
	SPI	MISO	GPIO19	Shared SPI
	SPI	SCK	GPIO18	Shared SPI
	GPIO	CE	GPIO4	
	GPIO	CSN	GPIO15	Dedicated Chip Select
	Power	VCC	3.3V ONLY	Never connect to 5V
	Power	GND	GND	
	Optional	IRQ	NC	Not connected

Controller BOM & Pinout

Component	Interface	Signal	ESP32 GPIO	Notes
HX711	GPIO	DT (DOUT)	GPIO34	Input only
	GPIO	SCK	GPIO25	
	Power	VCC	5V / 3.3V*	Module dependent
	Power	GND	GND	
SH1106 OLED	I²C	SDA	GPIO21	Address 0x3C
	I²C	SCL	GPIO22	
	Power	VCC	3.3V / 5V	Depends on module
	Power	GND	GND	
4×4 Keypad	Matrix	Row0	GPIO32	
		Row1	GPIO33	
		Row2	GPIO26	
		Row3	GPIO27	
		Col0	GPIO12	Strapping pin
		Col1	GPIO13	
		Col2	GPIO14	
		Col3	GPIO15	Strapping pin
nRF24L01+	SPI	MOSI	GPIO23	Shared SPI
	SPI	MISO	GPIO19	
	SPI	SCK	GPIO18	
	GPIO	CE	GPIO16	
	GPIO	CSN	GPIO17	
	Power	VCC	3.3V ONLY	
	Power	GND	GND	

Options

1. ADS1232 (Alternative ADC)

Instead of HX711

Signal	GPIO
DOUT	GPIO34
SCLK	GPIO25
PDWN	GPIO26
SPEED	GPIO27
A0	GPIO14

Can’t be used with 4x4 keypad without pc8574

2. MAX7219 7-Segment Display (Optional Display)

Instead of the OLED.

Signal	GPIO
DIN	GPIO23
CLK	GPIO18
CS	GPIO5
VCC	5V
GND	GND

Shares the SPI bus with the nRF24.

3. Multiple Load Cells

The firmware supports:

- **1 Load Cell (current)**
- **2 Load Cells**
- **4 Load Cells**

For multiple load cells, each requires its own dedicated ADC channel/pin set, and the arrays in pins.h must be expanded accordingly.